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Nowcasting economic activity in European regions using a mixed-frequency dynamic factor model☆

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ABSTRACT

Timely and accurate information about regional economic conditions can be essential for planning, implementing, and evaluating locally targeted economic policies. However, European regional accounts for output are published at an annual frequency with a two-year delay. To obtain robust and more timely measures in a computationally efficient manner, we propose a mixed-frequency dynamic factor model that accounts for national information to produce high-frequency estimates and reliable nowcasts of regional gross value added (GVA) in more than 150 regions across 13 European countries.

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1. Introduction

The timely availability of information on the state of regional economies is essential for the implementation and evaluation of targeted economic policies. These initiatives are often designed to address specific economic challenges in a particular region, and their effectiveness can only be accurately assessed if policymakers have access to timely, detailed data on the prevailing economic conditions in that region. However, EU-specific data collected by Eurostat on regional output and income comes with three limitations: (i) the frequency of publication is annual, (ii) consistent and homogeneous coverage of regions has only become available from the early 2000s onwards, and (iii) new data is released with a significant delay of two years (European Commission and Eurostat, 2023). These shortcomings, combined with cross-country and cross-regional heterogeneity in economic fundamentals, severely complicate the monitoring and fine-tuning of policy adjustments.

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In this paper, we aim to address the issue of limited and delayed regional economic data by leveraging more frequently available national data. We propose a mixed-frequency dynamic factor model (MF-DFM) that utilizes national information, available quarterly, to obtain quarterly estimates of regional gross value added (GVA), which is only available at an annual frequency and with a substantial time lag. Different to earlier studies—which use versions of MF-DFMs to extract information from a large panel of higher frequency series to predict one or more low frequency variables (see, e.g., Antolín-Díaz, Drechsel, & Petrella, 2024)—we deploy the factor structure as a tool to efficiently model a cross-section of many regions within a given country, with access to only very few observations over time.

The need to combine low- and high-frequency information in multivariate time series models has a long precedent in the literature (see, e.g., Frale, Marcellino, Mazzi, & Proietti, 2011; Grassi, Proietti, Frale, Marcellino, & Mazzi, 2015; Huber, Koop, Onorante, Pfarrhofer, & Schreiner, 2023; Mariano & Murasawa, 2003, 2010; Proietti & Moauro, 2006; Schorfheide & Song, 2015, 2020). These models are often used to obtain a monthly (or higher) frequency estimate of quantities of interest (such as GDP or other indicators of economic activity), using information from higher frequency time series such as industrial production, unemployment rates, or financial data. By contrast, our goal is to obtain estimates of *regional* GVA at a *quarterly* frequency. We follow the Nomenclature of Territorial Units for Statistics (NUTS) by Eurostat and focus on NUTS2-level regions, implying that the dimension of the cross-section becomes sizeable.[1] This calls for a model that is capable of handling a large number of time series simultaneously. The potentially large size of the cross-section is one problem; the annual frequency of our target variables is another. In our empirical work, we model quarterly movements in annual regional data dating back to only 2001—considering the required lag-structures, data availability results in 23 observations over time for each region (in the full sample), and even shorter series when we conduct out-of-sample forecast experiments.

In a recent paper, Koop, McIntyre, Mitchell, and Poon (2020) use a model for regional GVA in the United Kingdom (UK). This model includes a moderate number of time series (the 12 NUTS1 regions in the UK and a few additional national quantities with annual data dating back to the 1970s) and applies Markov Chain Monte Carlo (MCMC) estimation techniques to carry out estimation and inference. Additionally, they impose cross-sectional restrictions, ensuring that the sum of the regional GVA series (at the quarterly level) equals the national aggregate. Additional work and refinements are provided in Koop, McIntyre, and Mitchell (2019), Koop, McIntyre, Mitchell, and Poon (2024a), Koop, McIntyre, Mitchell, Poon, and Wu (2024b).

Other related papers compile datasets (see Bokun, Jackson, Kliesen, & Owyang, 2023) and design econometric frameworks for use with US state-level data. For instance, Hamilton and Owyang (2012) trace regional recessions, and Baumeister, Leiva-León, and Sims (2024) estimate weekly indicators of economic activity across US states using weekly, monthly, and quarterly information. Camacho and Leiva-Leon (2019) and Agudze, Billio, Casarin, and Ravazzolo (2022) are also examples of works investigating the US regional business cycle. Another strand of the literature uses panel methods to model regional data. For instance, Baltagi, Fingleton, and Pirotte (2014) study the predictive performance of a dynamic spatial panel data model with spatially correlated innovations, applied to forecast regional GVA in Europe. Other work at a comparable level of regional granularity often zooms in on specific countries or even subsets of regions within countries, e.g., as Claudio, Heinisch, and Holtemöller (2020), Henzel, Lehmann, and Wohlrabe (2015), Lehmann and Wikman (2025), Lehmann and Wohlrabe (2015) for Germany, or Billé, Tomelleri, and Ravazzolo (2023) for Italy.

We contribute econometrically and empirically along two key dimensions. First, we work on a granular level of geographical aggregation and model regional NUTS2 European data on a country-by-country basis. This results in a large cross-section of regions (in our empirical work, we have more than 150 regions). Second, data at this level of disaggregation for the EU are scarce and gathered at very low frequencies. We propose a DFM that combines the virtues of the MF-DFM proposed in Mariano and Murasawa (2010) with the cross-sectional and intertemporal restrictions of Koop et al. (2020). We assume that the latent high-frequency quantities of regional GVA co-move and feature a factor structure. This choice is motivated by the fact that regions close to each other (in geographical or economic terms) are often exposed to common economic shocks, which might trigger substantial co-movements across regional output dynamics.

A factor model is capable of handling such behaviour in a very parsimonious manner. Since the model may be prone to overfitting if the number of factors and/or lags is large, particularly given our short time series, we employ Bayesian shrinkage priors on the VAR coefficients to determine the law of motion of the high-frequency measures as well as the factor loadings. The resulting model can be applied to large cross-sections, as it effectively addresses computational and statistical issues by reducing the state space through factor restrictions and the introduction of highly flexible shrinkage priors.[2]

Our resulting regional MF-DFM is then applied country-by-country to model the corresponding NUTS2 regions within a particular country. We first provide predictive

[1] The NUTS geographical nomenclature divides the European economic territory into regions at three different levels. Following Eurostat definitions, NUTS2 are basic regions, while NUTS1 are major socio-economic regions. More details are available at ec.europa.eu/eurostat/web/nuts.

[2] A notable exception with respect to our comments about computational limitations is Chan, Poon, and Zhu (2023), who discuss efficient algorithms for estimating high-dimensional state space models. A possible solution to scalability issues is also provided in Gefang, Koop, and Poon (2020). This paper uses variational Bayes (VB) approximation techniques to estimate mixed-frequency VARs. However, the resulting model can be difficult to tune, and the size of the approximation errors induced by the VB-based estimation can be difficult to assess.

evidence that our model improves the nowcasting performance, often at sizeable margins, and in virtually all regions. While we find that the model improves forecasts across all considered countries, there is some heterogeneity in predictive performance. Based on this forecast exercise, we discuss several modelling aspects that are relevant for practitioners. Finally, we investigate high-frequency dynamics of the resulting quarterly regional time series and provide a brief (regional) case study about the onset of the COVID-19 pandemic.

The remainder of the paper is structured as follows. Section 2 discusses the econometric framework and implementation. Section 3 introduces the dataset and presents our forecast exercise, along with additional empirical results. The final section concludes and lists some areas for future research.

2. Econometric framework

2.1. The mixed-frequency dynamic factor model

For modelling economic activity across disaggregate regions, we propose an MF-DFM. Our model exploits the fact that regions within a country tend to co-move due to their exposure to common business cycle dynamics and shocks. Let $\boldsymbol{y}_t^* = (y_{1t}^*, \dots, y_{Nt}^*)'$ denote a vector of observed real GVA growth for regions $i = 1, \dots, N$, where $t = 1, \dots, T$, runs on a quarterly frequency.[3] This implies that $\boldsymbol{y}_t^*$ features missing values, since regional GVA is only observed annually and with a lag. We model these quantities as year-on-year differences, such that our baseline dataset is approximately stationary, a standard transformation when using DFMs with macroeconomic data (see, e.g., McCracken & Ng, 2016; Stock & Watson, 2002). Further, we refer to the never observed (latent quarterly) GVA growth by region as $\boldsymbol{y}_t = (y_{1t}, \dots, y_{Nt})'$.

Define $\tilde{\boldsymbol{f}}_t$ as a Q_f-vector of latent factors, and $\boldsymbol{z}_t$ are Q_z-observed factors. We collect these in the $Q(= Q_f + Q_z) \times 1$ vector $\boldsymbol{f}_t = (\tilde{\boldsymbol{f}}_t', \boldsymbol{z}_t')'$. That is, typically $Q \ll N$, and the framework we propose thus only relies on modelling Q_f latent processes that govern within-year dynamics across regions, rather than N separate processes (individually for each region). The factors $\tilde{\boldsymbol{f}}_t$ are never observed. The vector $\boldsymbol{z}_t$ is observed at a quarterly frequency and may include additional information, such as interest rates, unemployment rates, or other national indicators that can affect economic activity across regions.[4] In addition, $z_{\text{GVA},t}$ denotes quarter-on-quarter log-differences of GVA on the national level. This quantity is observed each quarter, does *not* enter the vector $\boldsymbol{z}_t$, and is used in the cross-sectional restriction (2) to impose that GVAs of the regions are compatible with the national one.

We link observed regional GVA measures to the factors by combining an intertemporal restriction with a conventional measurement equation for factor models, as seen also in Mariano and Murasawa (2003, 2010). For quarters when we observe annual regional GVA growth ($t \in \mathcal{T}_{\text{obs}}$ for the set $\mathcal{T}_{\text{obs}} = \{4, 8, 12, \dots, T-4, T\}$), we have:

$$y_{it}^* = \sum_{q=1}^{Q_f} \lambda_{iq} \left(\frac{1}{4}\tilde{f}_{qt} + \frac{2}{4}\tilde{f}_{qt-1} + \frac{3}{4}\tilde{f}_{qt-2} + \tilde{f}_{qt-3} + \frac{3}{4}\tilde{f}_{qt-4} + \frac{2}{4}\tilde{f}_{qt-5} + \frac{1}{4}\tilde{f}_{qt-6} \right) + \epsilon_{it}, \tag{1}$$

with $\epsilon_{it} \sim \mathcal{N}\left(0, \sigma^2_{\text{ME},i}\right)$, while in the quarters without regional observations $y_{it}^* = \varnothing$ (when $t \notin \mathcal{T}_{\text{obs}}$).[5] This equation serves two main purposes. First, it distributes the latent growth rates of the quarter throughout the year via the triangular weighting scheme, which arises from considering an approximation to log-growth rates, as proposed in Mariano and Murasawa (2003). Second, the unobserved and observed latent factors are then linked to observed regional growth by the factor loadings λ_{iq}. In related papers, the intertemporal restriction is typically specified to hold exactly,[6] but due to our factor structure we introduce region-specific Gaussian measurement errors centered on zero with variance $\sigma^2_{\text{ME},i}$.

For later reference, we store the loadings for regions $i = 1, \dots, N$, and factor $q = 1, \dots, Q$, in an $N \times Q$ matrix $\boldsymbol{\Lambda}$, which has its last Q_z columns equal to $\boldsymbol{0}_{N \times Q_z}$. In terms of the never observed latent quarterly regional GVA, we may write our specification statically as $\boldsymbol{y}_t = \boldsymbol{\Lambda}\boldsymbol{f}_t + \boldsymbol{\epsilon}_t$ where $\boldsymbol{\epsilon}_t = (\epsilon_{1t}, \dots, \epsilon_{Nt})'$. Let $\boldsymbol{\lambda}_i$ denote the ith row of the factor loadings $\boldsymbol{\Lambda}$, then $y_{it} = \boldsymbol{\lambda}_i'\boldsymbol{f}_t + \epsilon_{it}$. Here, it is worth noting that our framework does not distinguish between the different potential sources of variance, e.g., whether the variance is due to the approximate nature of the intertemporal restriction or regional idiosyncrasies that deviate from national dynamics, contributing to measurement errors. See also our discussion in the next sub-section in this context.

Inspired by Koop et al. (2020)—and since the sum of regional GVA will approximately equal national GVA in *levels*—we may aggregate the latent regional observations. Note that Koop et al. (2020) directly model y_{it} without a factor structure and thus do not need an additional mapping via the factor loadings to the space of observables. This step is therefore one of the distinguishing features of our proposed approach. Indeed, a key feature of our setup is that regional growth rates can still be linked

[3] In our empirical application, we estimate the regional MF-DFM country-by-country, which is why we omit a dedicated country-index to keep notation simple. A joint model of all EU regions is a possible extension of our proposed framework. We note, however, that this would imply having at least three distinct levels of aggregation and thus require a significantly more complicated factor structure, alongside a system of intricate cross-sectional restrictions. We leave this option for future research.

[4] This feature offers additional inferential possibilities such as identifying structural shocks nationally, and tracing their propagation on a regional level. We leave such aspects for future research.

[5] We also have identities as measurement equations when observed factors are included, see Section 2.3.

[6] Chan et al. (2023) discuss "soft constraints" as a tool to address potential approximation errors due to the construction of the intertemporal aggregation, and as a means to increase computational efficiency in high-dimensional models by introducing a very small amount of variance. Typically, these restrictions are expected to hold almost exactly, however, so the measurement error in our case primarily is introduced due to using a lower dimensional set of factors rather than modeling an individual latent processes by region.

to those observed on the national level, quarterly. Note that in expectations (or when the measurement errors are very small, as is to be expected in our context), for the unobserved quarterly regional GVA, we have $\mathbb{E}(y_{it}) = \boldsymbol{\lambda}_i' \boldsymbol{f}_t = \sum_{q=1}^{Q} \lambda_{iq} f_{qt}$.

Due to using *growth rates*,[7] and since in our case regions may be severely imbalanced with respect to their size, we rely on the proportional weighting scheme usually associated with exact growth rates, i.e., the national growth rate is the weighted average of regional growth rates:

$$\sum_{i=1}^{N} \omega_i \mathbb{E}(y_{it}) = \sum_{i=1}^{N} \omega_i \sum_{q=1}^{Q} \lambda_{iq} f_{qt} = \boldsymbol{\omega}' \boldsymbol{\Lambda} \boldsymbol{f}_t,$$

where $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_N)'$ are pre-determined weights that reflect the share of the respective region in national GVA.[8] We may impose this restriction for all $t = 1, \ldots, T$, with the following additional measurement equation, which we label the cross-sectional restriction:

$$z_{\text{GVA},t} = \boldsymbol{\omega}' \boldsymbol{\Lambda} \boldsymbol{f}_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}\left(0, \sigma_{\text{CS}}^2\right). \tag{2}$$

This cross-sectional restriction does not necessarily hold exactly, for various reasons. First, such an aggregation scheme is typically only approximate in nature (see also Koop et al., 2020; Mitchell, Smith, Weale, Wright, & Salazar, 2005). Second, while regional accounts use the same concepts as national accounts, there are conceptual and measurement limitations at the regional level.[9] Third, and related to the previous point, usually not all parts of national output can be uniquely associated with a region. For instance, in the UK or US, national measures comprise overseas activity (e.g., oil and gas output from elsewhere, or the military, see Koop et al., 2020; Koop, McIntyre, Mitchell, & Raftapostolos, 2025). In the European context, there are both overseas territories and extra-regional entities that exhibit unusual time series properties (although they are typically not of significant economic importance) or are generally not part of the rest of the NUTS classification. These aspects are reflected in the cross-sectional aggregation error ε_t.

Mappings of the latent states. Several aspects about how the latent and observed quantities may be processed in our modeling framework are worth noting. As discussed above, one direct output of our model is the full history of the quarterly latent and observed factors, $\{\boldsymbol{f}_t\}_{t=1}^{T}$, and associated never observed regional quarterly GVA, $\{\boldsymbol{y}_t\}_{t=1}^{T}$. The latter can be obtained by projecting the factors to the original space of the regions via the factor loadings and then adding the measurement error.

An additional quantity, which we label the annualized factors (indicated with superscript "a"), can be obtained by applying the intertemporal restriction to the quarterly latent factors:

$$\tilde{f}_{qt}^{(a)} = \frac{1}{4}\tilde{f}_{qt} + \frac{2}{4}\tilde{f}_{qt-1} + \frac{3}{4}\tilde{f}_{qt-2} + \tilde{f}_{qt-3} + \frac{3}{4}\tilde{f}_{qt-4} + \frac{2}{4}\tilde{f}_{qt-5} + \frac{1}{4}\tilde{f}_{qt-6},$$

for $t = 1, \ldots, T$, which we use to define $\boldsymbol{f}_t^{(a)} = (\tilde{f}_{1t}^{(a)}, \ldots, \tilde{f}_{Qt}^{(a)})'$. These are not of primary interest. But a projection via the factor loadings and adding the measurement error yields $\boldsymbol{y}_t^{(a)} = \boldsymbol{\Lambda} \boldsymbol{f}_t^{(a)} + \boldsymbol{\epsilon}_t$, the annualized (quarterly) regional GVA, which can be computed in each quarter t. Note that for $t \in \mathcal{T}_{\text{obs}}$, (1) states that $\boldsymbol{y}_t^{(a)} = \boldsymbol{y}_t^*$, which also defines the targeted quantity that we can use for evaluating our predictions with observed realized values from a holdout sample in Section 3 (i.e., the final quarter, in terms of $\boldsymbol{y}_t^{(a)}$, is the annual nowcast).

Dynamics of the states. The final aspect of our framework refers to how we model the dynamics of the observed and latent states. We do so by using a vector autoregressive (VAR) state equation with $P \geq 7$ lags (this number is due to the annual/quarterly frequency mismatch and the intertemporal restriction given in (1), from an implementation perspective):

$$\boldsymbol{f}_t = \boldsymbol{A}_1 \boldsymbol{f}_{t-1} + \ldots + \boldsymbol{A}_P \boldsymbol{f}_{t-P} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \overset{\text{iid}}{\sim} \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Omega}_t). \tag{3}$$

Here, $\boldsymbol{A}_p$ are $Q \times Q$-matrices of VAR coefficients associated with the respective pth lag, and $\boldsymbol{u}_t$ is a Gaussian error term with zero mean and (potentially time-varying) $Q \times Q$-covariance matrix $\boldsymbol{\Omega}_t = \boldsymbol{B}_0^{-1} \boldsymbol{H}_t \boldsymbol{B}_0^{-1'}$, where $\boldsymbol{B}_0^{-1}$ is a lower triangular matrix with unit diagonal and $\boldsymbol{H}_t = \text{diag}(h_{1t}, \ldots, h_{Qt})$ is a diagonal matrix collecting the variances of independent shocks. Our measurement equations, in conjunction with this state equation, yield a fairly straightforward Gaussian state-space model.

For estimation, it is convenient to cast the model in its structural form as:

$$\boldsymbol{B}_0 \boldsymbol{f}_t = \boldsymbol{B}_1 \boldsymbol{f}_{t-1} + \ldots + \boldsymbol{B}_P \boldsymbol{f}_{t-P} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{H}_t),$$

where the reduced form coefficients can be recovered from computing $\boldsymbol{A}_p = \boldsymbol{B}_0^{-1} \boldsymbol{B}_p$ for $p = 1, \ldots, P$, $\boldsymbol{B}_0^{-1} \boldsymbol{\eta}_t = \boldsymbol{u}_t$ and $\boldsymbol{\eta}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{H}_t)$. We will use this form to specify our priors and to design an efficient posterior simulation algorithm.

2.2. Prior setup

Our model is simpler than larger VAR models that explicitly feature equations for each region due to the proposed factor structure. However, the size of the cross-section and the required number of factors still yield a large number of parameters to be estimated. The latter also implies that (3) may result in a comparatively large VAR. We thus regularise the parameter space via shrinkage priors.

Our prior setup exploits the favourable shrinkage properties of a global-local prior (see, e.g., Huber & Feldkircher, 2019, for a discussion). Specifically, we shrink

[7] Koop et al. (2020) also model log-growth rates, but use an unweighted average to define their restriction. They show this to be a useful approximation when using log-growth rates, instead of the required proportional weighting that arises when using exact growth rates.

[8] The shares in $\boldsymbol{\omega}$ may, in principle, vary over time. However, in our empirical application (and likely most cases where our framework will be applied), they are virtually constant, and we omit time indexing here.

[9] See ec.europa.eu/eurostat/web/national-accounts for more details about the respective methodologies.

the $\boldsymbol{B}_p$'s, and the free elements of $\boldsymbol{B}_0$ and $\boldsymbol{\Lambda}$, via horseshoe (HS, see Carvalho, Polson, & Scott, 2010) priors. Let $\boldsymbol{b} = \text{vec}(\boldsymbol{B}_1, \ldots, \boldsymbol{B}_p)$ denote the $k(= PQ^2)$−dimensional vector of VAR coefficients, $\boldsymbol{b}_0$ the $m(= Q(Q-1)/2)$−dimensional vector that stores the free elements in $\boldsymbol{B}_0$ and $\text{vec}(\boldsymbol{\Lambda})$ are the vectorized factor loadings. Our prior is then given by:

$$p(\boldsymbol{b}) \propto \prod_{j=1}^{k} \mathcal{N}(b_j|0, \tau_{b,j}^2 \xi_b) \times \mathbb{I}(\boldsymbol{b} \in \mathcal{S}),$$

$$p(\boldsymbol{b}_0) = \prod_{j=1}^{m} \mathcal{N}(b_{j,0}|0, \tau_{b_0,j}^2 \xi_{b_0}),$$

$$p(\text{vec}(\boldsymbol{\Lambda})) = \prod_{j=1}^{QN} \mathcal{N}(\lambda_j|0, \tau_{\lambda,j}^2 \xi_\lambda),$$

with b_j, $b_{0,j}$ and λ_j denoting the corresponding jth element of $\boldsymbol{b}$, $\boldsymbol{b}_0$, $\text{vec}(\boldsymbol{\Lambda})$, respectively. We let the set $\mathcal{S}$ define the stationary region of the reduced form model in (3) so that the indicator function $\mathbb{I}$ equals 1 if $\boldsymbol{b}$ is within the stationary region.[10] The hyperparameters of the priors are $\tau_{s,j}^2$ and ξ_s for $s \in \{b, b_0, \lambda\}$. The local shrinkage parameter $\tau_{s,j}^2$ allows for coefficient-specific shrinkage towards the prior mean, while the global shrinkage factor ξ_s pulls all elements in a particular vector towards the prior mean. To obtain the horseshoe, we use Half-Cauchy priors on both $\tau_{s,j}$ and ξ_s.

The remaining priors relate to innovations in the state and measurement equations. When we consider homoskedastic variances in the state equation (i.e., when $\boldsymbol{H}_t = \boldsymbol{H}$ for all t), we set a moderately informative inverse Gamma prior, $h_q \sim \mathcal{G}^{-1}(3, 0.3)$ for $q = 1, \ldots, Q$. In case we consider heteroskedastic errors, we rely on the following independent AR(1) processes, $\log(h_{qt}) = \mu_{hq} + \phi_{hq}(\log(h_{qt-1}) - \mu_{hq}) + \varsigma_{hq} e_{qt}$, with $e_{qt} \sim \mathcal{N}(0, 1)$, that yield an otherwise standard stochastic volatility model that we estimate with the prior setup and algorithm described in Kastner and Frühwirth-Schnatter (2014).

Regarding measurement errors, we impose additional prior information. Since our intertemporal restriction is combined with a factor model, it is not exactly binding as in related papers. However, as we expect regional business cycles to be relatively synchronized, our prior intuition is that these errors will typically be small. To assess the impacts of these priors, we experiment with various possible choices of informative priors. The tightest variant of this prior, which we indicate as `me:tight` assumes $\sigma_{\text{ME},i}^2 \sim \mathcal{G}^{-1}(1000, 0.01)$ for $i = 1, \ldots, N$, while a looser variant, `me:loose`, has $\sigma_{\text{ME},i}^2 \sim \mathcal{G}^{-1}(100, 0.01)$. These choices express the belief that the restriction almost holds, although the prior allows for deviations in case it is required due to regional heterogeneity. Similarly, we try tighter and looser variants on the variance of the cross-sectional restriction. The first, `csr:tight`, is achieved with $\sigma_{\text{CS}}^2 \sim \mathcal{G}^{-1}(500, 0.05)$, while the latter, `csr:loose`, assumes $\sigma_{\text{CS}}^2 \sim \mathcal{G}^{-1}(50, 0.05)$. The loosest variant, in some sense, is achieved by deleting the cross-sectional measurement equation (`csr:no`) altogether.

[10] We implement this via rejection sampling, but note that empirically the number of rejected draws is low.

2.3. An overview of the posterior simulation algorithm

Depending on the number of factors and lags to include, the model in (3) can become enormous and, despite the factor structure, computation could become unfeasible when the state equation is treated as a full system. Additionally, this setup leaves the option open to add more flexible conditional mean or variance specifications. Hence, we employ the structural form of the model and conduct estimation on an equation-by-equation basis.

Using $\tilde{\boldsymbol{B}}_0 = \boldsymbol{B}_0 - \boldsymbol{I}_Q$, we may write the VAR as:

$$\boldsymbol{f}_t = \boldsymbol{B}_1 \boldsymbol{f}_{t-1} + \ldots + \boldsymbol{B}_P \boldsymbol{f}_{t-P} + \tilde{\boldsymbol{B}}_0(-\boldsymbol{f}_t) + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{H}_t).$$

Further, for $q = 2, \ldots, Q$, let $\boldsymbol{x}_{qt} = (\boldsymbol{f}_{t-1}', \ldots, \boldsymbol{f}_{t-P}', -f_{1t}, \ldots, -f_{q-1t})$ with associated parameters $\boldsymbol{\beta} = (\boldsymbol{B}_{1[q,\bullet]}, \ldots, \boldsymbol{B}_{P[q,\bullet]}, \tilde{\boldsymbol{B}}_{0[q,1]}, \ldots, \tilde{\boldsymbol{B}}_{0[q,q-1]})'$ where the squared brackets index the elements of the respective matrix. Conditionally, we thus obtain the independent linear regressions:

$$f_{qt} = \boldsymbol{x}_{qt}' \boldsymbol{\beta} + \eta_{qt}, \quad \mathcal{N}(0, h_{qt}),$$

and the posterior of $\boldsymbol{\beta}$ takes a textbook Gaussian form, and the variances h_{qt} can also be sampled equation-by-equation using standard posteriors and algorithms. These draws can then be used to recover the reduced form parameters of (3).

Define $\boldsymbol{s}_t = (\boldsymbol{f}_t', \boldsymbol{f}_{t-1}', \boldsymbol{f}_{t-2}', \boldsymbol{f}_{t-3}', \boldsymbol{f}_{t-4}', \boldsymbol{f}_{t-5}', \boldsymbol{f}_{t-6}')'$ and let $\boldsymbol{\iota}_N$ be an $N \times 1$-vector of 1s. Our restrictions in stacked notation yield measurement equations with time-varying dimensions (and parameters) of the form:

$$\tilde{\boldsymbol{y}}_t^* = \boldsymbol{L}_t \boldsymbol{s}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{\Sigma}_t). \tag{4}$$

Indeed, the loadings $\boldsymbol{L}_t$ and covariance matrix $\boldsymbol{\Sigma}_t$ vary over time because the dimension of the measurement equation changes. We distinguish between two sets of parameters, as described in Section 2.1: when the annual regional GVA is observed, and when it is not. In the former case, we have $\tilde{\boldsymbol{y}}_t^* = (\boldsymbol{y}_t^{*'}, z_{\text{GVA},t}, \boldsymbol{z}_t')'$, whereas in the latter, $\tilde{\boldsymbol{y}}_t^* = (z_{\text{GVA},t}, \boldsymbol{z}_t')'$. $\boldsymbol{L}_t$ is a combination of the loadings matrix $\boldsymbol{\Lambda}$, the intertemporal loadings in $\boldsymbol{D} = ((1/4) \cdot \boldsymbol{\iota}_{Q_f}', \boldsymbol{\iota}_{Q_z}', (2/4) \cdot \boldsymbol{\iota}_{Q_f}', \boldsymbol{0}_{Q_z}', (3/4) \cdot \boldsymbol{\iota}_{Q_f}', \boldsymbol{0}_{Q_z}', \ldots)$ and $\boldsymbol{\omega}$. Let $o_t = \mathbb{I}(t \in \mathcal{T}_{\text{obs}})$ be an indicator that is equal to 1 if regional GVA is observed and 0 otherwise, then:

$$\boldsymbol{L}_t = o_t \cdot \begin{bmatrix} [(\boldsymbol{\iota}_P' \otimes \boldsymbol{\Lambda}) \odot (\boldsymbol{\iota}_N \otimes \boldsymbol{D})] \\ \tilde{\boldsymbol{L}} \end{bmatrix} + (1 - o_t) \cdot \tilde{\boldsymbol{L}},$$

$$\tilde{\boldsymbol{L}} = \begin{bmatrix} \boldsymbol{\omega}' \boldsymbol{\Lambda} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}_{Q_z} & \boldsymbol{0} \end{bmatrix},$$

and $\boldsymbol{\Sigma}_t = o_t \cdot \text{diag}(\{\sigma_{\text{ME},i}^2\}_{i=1}^N, \boldsymbol{0}_{Q_z}', \sigma_{\text{CS}}^2) + (1 - o_t) \cdot \text{diag}(\boldsymbol{0}_{Q_z}', \sigma_{\text{CS}}^2)$. Combining the state equation given in (3) in companion form with (4) yields an otherwise conventional state space model and allows the use of standard state space methods. We follow Koop et al. (2020) and use a standard filtering/smoothing algorithm to obtain a draw from the full history of the latent states represented by the factors $\{\boldsymbol{f}_t\}_{t=1}^T$.

Conditional on a draw of the factors, we may update the loadings and measurement errors region-by-region using (1), and the variance of the cross-sectional restriction using (2), where all posteriors again take textbook forms based on the Bayesian linear regression model.

This enables a straightforward MCMC algorithm using Gibbs sampling, which iterates between known conditional posterior distributions to obtain draws from the joint posterior.

3. Nowcasting GVA in European regions

3.1. Data overview, nowcasting design and model specification

We study Euro area countries that have at least two NUTS2 regions and exclude all extra-region entities indexed by the ZZ suffix, which is used to indicate an area that is not part of the rest of the NUTS classification. Our multi-country data set consists of Austria (AT), Belgium (BE), Germany (DE), Greece (EL), Spain (ES), Finland (FI), France (FR, excluding overseas territories), Ireland (IE), Italy (IT), Lithuania (LT), the Netherlands (NL), Slovenia (SI) and Slovakia (SK).[11]

Like in Koop et al. (2020), our target variable is GVA rather than GDP, because GVA provides a consistent measure of economic growth between regions. In fact, GVA plus taxes and less subsidies on products equals GDP. We do not consider any additional observed factors per quarter; therefore, $Q_z = 0$. We obtain regional and national GVA figures from the Eurostat regional economic accounts. Our final data set comprises annual GVA for 161 NUTS2 regions in 13 European countries that start in 2001 and are released between February 15 and March 31 of each year, with a two-year publication delay.[12]

In our empirical work, we have three goals. First, we demonstrate that our model nowcasts yearly GVA figures with accuracy relative to a suitable benchmark (discussed below). Second, we zoom in on the differences between regions and between countries in the nowcast performance and tease out the predictive properties of our model. Third, we discuss the qualitative features of the quarterly regional nowcasts for selected periods, regions, and countries.

To produce and evaluate our pseudo-real-time nowcasts, we rely on the following nowcasting design. We want to nowcast regional GVA growth in the year ending at time T, where $t = 1, \ldots, T$, runs on a quarterly frequency. We assume that our nowcast is produced during the first quarter of the following year ($T + 1$), when all quarters of national quarterly output for the target year have been released. To account for the two-year publication lag in the regional data, we run the model using annual regional data until $T - 8$ and quarterly national data observed up to the end of year T. The annual value that we can use to evaluate our predictions with ex post realizations is thus $y_{iT}^{(a)}$ for the nowcast, and the backcast is $y_{iT-4}^{(a)}$, see also Section 2.1 for details about the required intertemporal transformations. This information set reflects the actual release schedule for yearly and regional data, as well as the associated publication delays. In our empirical application, we produce out-of-sample predictions of annual regional GVA growth from 2016 to 2023 every year. Given the short time sample available, we adopt an expanding window approach. Crucially for this exercise, we respect the release calendar, but note that we do not explicitly investigate the role of potential ex-post data revisions.

For each country in our data set, we begin by estimating the MF-DFM independently and select the optimal number of factors, Q, based on their density nowcasting performance. We use a grid of values for factors ranging from 1 to the minimum between the number of regions in a country and an experimental threshold $Q_{\max} = 12$. In other words, for countries with fewer than 12 NUTS2 regions, we test a "fully-specified" model, while for the other countries we take 12 as the maximum number of factors. Note that the overspecified factor model closely resembles the MF-VAR model by Koop et al. (2020).

In principle, modeling all countries (and associated regions) within a single model would be possible. However, we opt not to do so for three reasons. First, while still being tractable, computation becomes much more cumbersome. This is because the space of observed series becomes significantly larger, and filtering algorithms tend to be slow. Moreover, since we may need to include a larger number of factors, the dimension of the state equation may also increase substantially. Second, model calibration becomes an issue since we introduce both cross-sectional and intertemporal restrictions. Third, we aimed to exploit cross-country heterogeneity in the number of regions to investigate its consequences for the choice of the number of endogenous factors.

We evaluate the nowcasting performance through the continuous rank probability score (CRPS, Gneiting & Raftery, 2007), which we weight by the historical regional GVA shares when the focus is on the aggregate predictive accuracy of the country. This provides an overall measure of regional density nowcasting performance, where regions are weighted by their relative economic significance.

As benchmark forecasting models, using the same information set as the MF-DFM, we have experimented with several autoregressive (AR) specifications at an annual frequency (i.e., $t = 4, 8, 12, \ldots, T - 8$) that are nested in the autoregressive specification $y_{it}^* = c_i + \phi_i y_{it-4}^* + e_{it}$ with $e_{it} \sim \mathcal{N}(0, s_i^2)$. All benchmarks are estimated with Bayesian methods under loose priors. A random walk (RW, in levels) arises for $c_i = \phi_i = 0$.[13] The most competitive benchmark model that we consider is the historical mean, which results in $\phi_i = 0$. Any relative predictive losses presented below are benchmarked against the historical mean as ratios. We note that our holdout consists of only 8 observations, and we thus refrain from formally testing for statistical significance of differences between competing approaches.

[11] We excluded Portugal from our analysis because the regional classification changed in this country between 2021 and 2024 (splits of regions and boundary changes). We were unable to obtain a consistent set of uniquely identifiable regions for the required period.

[12] See also ec.europa.eu/eurostat and ec.europa.eu/eurostat/cache/metadata/en/reg_eco10_esms.htm.

[13] The results for the RW and AR models are available upon request.

Table 1
Average (2016–2023) out-of-sample nowcast ($h = 0$) CRPS by country and number of factors, for MF-DFM variants. Region-specific CRPSs are weighted by the historical GVA share. CRPSs are shown as ratios to the historical mean nowcast (values below 1 signal improvement and vice versa). Specifications: homoskedastic (`hom`), stochastic volatility (`SV`); measurement error (`ME`) prior variance `tight`. Darker shades indicate stronger nowcasting gain.

		Factors (Q)	AT	BE	DE	EL	ES	FI	FR	IE	IT	LT	NL	SI	SK
MF-DFM-hom	CSR:tight	ME:loose	1	0.69	0.70	0.59	0.54	0.54	0.55	0.70	0.81	0.55	0.45	0.64	0.51	**0.48**
			2	0.67	0.69	0.58	0.53	0.53	0.53	0.66	1.02	0.50	0.41	0.60	0.49	0.49
			3	0.65	0.66	0.57	0.51	0.51	0.51	0.65	0.93	0.49		0.60		0.50
			4	0.66	0.65	0.56	0.50	0.50	0.51	0.62		0.47		0.61		0.49
			5	0.64	0.64	0.55	0.49	0.49	**0.51**	0.60		0.46		0.60		
			6	0.63	0.63	0.55	0.48	0.47		0.59		0.44		0.58		
			7	0.62	0.62	**0.54**	0.47	0.45		0.57		0.42		0.57		
			8	0.61	0.59	0.55	0.46	0.44		0.55		0.42		0.54		
			9	**0.59**	0.58	0.55	0.45	0.43		0.54		0.41		0.53		
			10		0.56	0.56	**0.44**	**0.42**		**0.52**		**0.41**		**0.52**		
			11		**0.56**	0.59	0.44	0.43		0.53		0.41		0.53		
			12			0.67	0.50	0.46		0.56		0.45		0.58		
		ME:tight	1	0.80	0.84	0.69	0.80	0.79	0.69	0.87	0.97	0.62	0.58	0.74	0.62	0.76
			2	0.77	0.74	0.60	0.51	0.60	0.58	0.75	1.04	0.60	0.51	0.67	0.60	0.49
			3	0.76	0.75	0.59	0.51	0.58	0.54	0.72	0.94	0.50		0.65		0.51
			4	0.76	0.76	0.60	0.50	0.57	0.55	0.70		0.47		0.65		0.53
			5	0.75	0.73	0.58	0.49	0.55	0.56	0.67		0.46		0.64		
			6	0.74	0.73	0.57	0.49	0.53		0.66		0.48		0.61		
			7	0.72	0.72	0.56	0.48	0.52		0.66		0.48		0.60		
			8	0.71	0.70	0.56	0.48	0.50		0.64		0.47		0.61		
			9	0.69	0.69	0.55	0.47	0.49		0.64		0.46		0.59		
			10		0.67	0.55	0.47	0.47		0.62		0.45		0.57		
			11		0.65	0.57	0.48	0.48		0.61		0.45		0.56		
			12			0.71	0.66	0.54		0.62		0.51		0.59		
MF-DFM-SV	CSR:tight	ME:loose	1	0.83	1.06	0.88	0.69	0.61	0.88	0.65	0.79	0.63	0.70	0.79	0.83	0.62
			2	0.72	0.82	0.58	0.54	0.53	0.73	0.79	0.94	0.50	**0.40**	0.86	**0.46**	0.56
			3	0.70	0.71	0.58	0.58	0.56	0.69	0.62	0.82	0.50		0.82		0.60
			4	0.67	0.78	0.62	0.55	0.54	0.70	0.62		0.50		0.84		0.56
			5	0.71	0.72	0.63	0.59	0.52	0.68	0.62		0.50		0.83		
			6	0.71	0.72	0.63	0.59	0.54		0.62		0.51		0.79		
			7	0.69	0.73	0.66	0.64	0.56		0.63		0.51		0.80		
			8	0.71	0.71	0.67	0.65	0.57		0.65		0.52		0.76		
			9	0.74	0.72	0.71	0.66	0.57		0.68		0.53		0.80		
			10		0.76	0.76	0.65	0.57		0.75		0.54		0.80		
			11		0.83	0.85	0.71	0.63		0.80		0.59		0.87		
			12			1.05	0.86	0.77		0.91		0.71		0.95		
		ME:tight	1	0.88	1.02	0.93	1.00	1.03	1.04	1.07	0.96	0.72	0.78	1.04	0.87	1.12
			2	0.72	0.70	0.59	0.78	0.52	0.65	0.71	0.76	0.53	0.46	0.58	0.70	0.69
			3	0.85	0.71	0.59	0.61	0.55	0.58	0.73	**0.75**	0.50		0.62		0.58
			4	0.86	0.89	0.60	0.56	0.52	0.64	0.61		0.52		0.62		0.60
			5	0.85	0.85	0.59	0.54	0.52	0.74	0.57		0.50		0.61		
			6	0.85	0.86	0.60	0.56	0.53		0.59		0.53		0.61		
			7	0.82	0.78	0.61	0.59	0.54		0.57		0.56		0.66		
			8	0.82	0.84	0.63	0.62	0.53		0.60		0.56		0.78		
			9	0.78	0.81	0.62	0.61	0.56		0.60		0.56		0.81		
			10		0.82	0.62	0.60	0.58		0.67		0.54		0.82		
			11		0.83	0.65	0.67	0.63		0.67		0.53		0.80		
			12			0.75	0.89	0.77		0.74		0.58		0.84		

3.2. Nowcasting results

We begin by discussing the findings of our nowcasting exercise, averaged over out-of-sample years (i.e., from 2016 to 2023), obtained from the expanding window design evaluation scheme described above. These are provided in Table 1, which reports the weighted-average by country (region-specific CRPSs are weighted by the historical GVA share) in the columns. The displayed values are ratios of the indicated specification to the historical mean nowcast benchmark. The rows indicate the number of factors Q, while the groupings refer to prior and model specification details of MF-DFM: homoskedastic (`hom`), stochastic volatility (`SV`); measurement error (`ME`) prior variance `tight` and `loose`; cross-sectional restriction (`CSR`) prior variance `tight`, `loose` and `no` when the restriction is deleted from the model (results for the latter appear in the Appendix, see also the discussion below). Missing values in the table indicate the countries that have fewer than 12 NUTS2 regions: for these countries, the upper threshold for Q is set equal to the number of regions.

The best performing specification is always among the models that impose a tight prior on the cross-sectional restriction, so Table 1 only shows the results for `CSR:tight`. Additional results for backcasts (labeled $h = -1$) and other cross-sectional restriction specifications are shown in Appendix A. We highlight in bold the model specification that achieves the largest gain with respect to the historical mean benchmark. Examining the different model specifications, we observe that in 10 out of the 13 countries, the best-performing model is obtained with the homoskedastic MF-DFM, featuring a loose prior variance on the measurement error. The only exceptions are Ireland, Lithuania, and Slovenia, where the best-performing model is the MF-DFM with stochastic volatility. In the first case, the tight prior variance is preferred, while in the latter two cases, the best performance is achieved with a loose prior on the measurement error.

At a general level, we observe that our model produces nowcasts that are more accurate than those of the benchmark for all countries under consideration. These gains vary across countries and a number of factors. For some countries and values of Q, improvements are substantial. For instance, in the case of Greece, Spain, Italy, and Lithuania, we observe gains of approximately 60%. In Austria, our model produces smaller relative gains with respect to the benchmark; nevertheless, the accuracy premia of around 40% are still very substantial.

When we relate the nowcasting performance within a particular country to the number of factors, we find a consistent pattern. When the number of factors is very low, the predictive performance of the models is generally weaker, indicating underfitting. One exception is Slovakia,

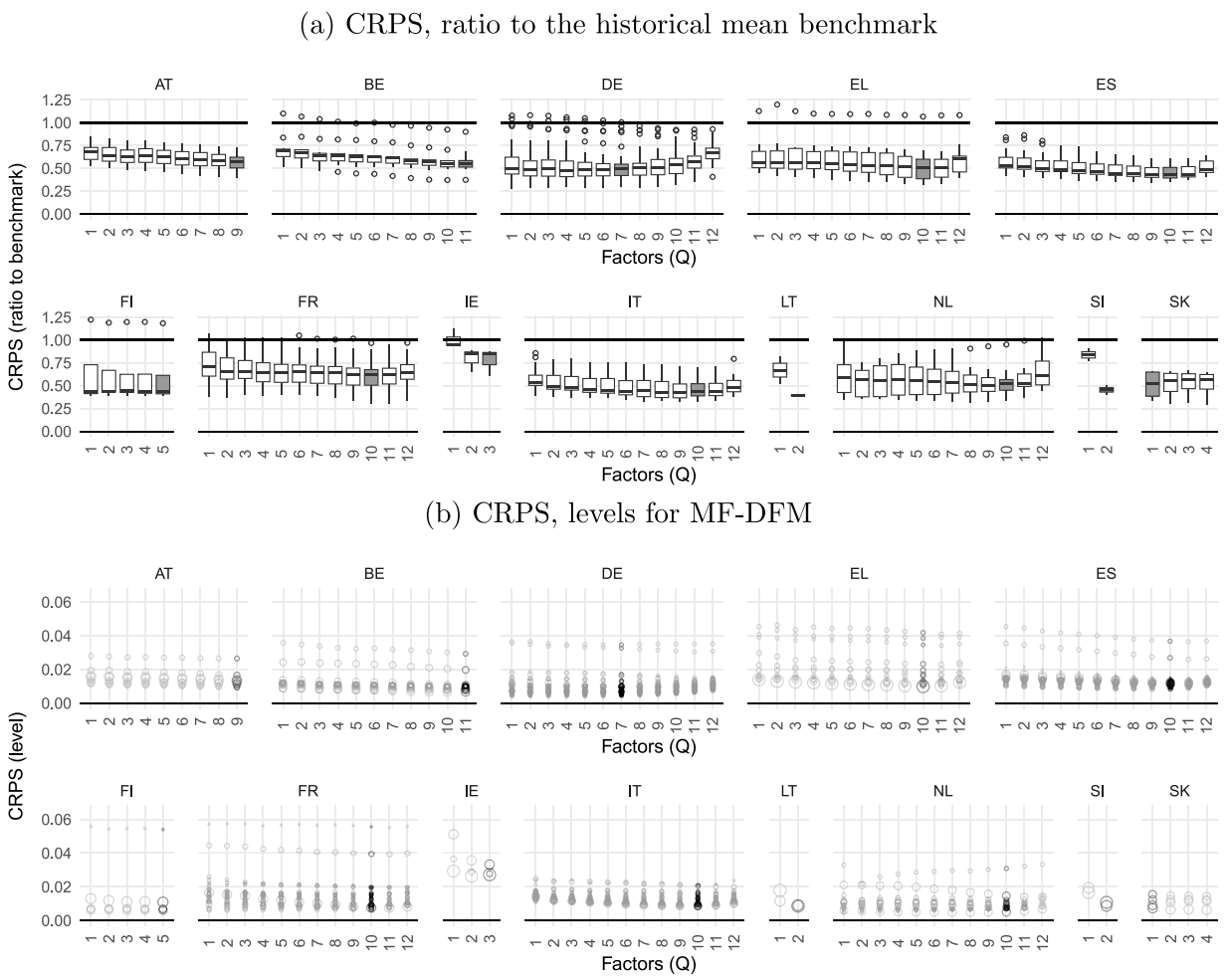


Fig. 1. Boxplots averaged over the holdout sample of relative regional CRPS across the number of factors for the best performing MF-DFM model (i.e., we fix the remaining specification details and vary only the number of factors) by country in the upper panel (a) and raw CRPS losses per region in the lower panel (b). The CRPSs in panel (a) are shown as ratios to the historical mean nowcast (values below 1 signal improvement and vice versa). The size of the circles in panel (b) indicates the relative weight of the respective region in national GVA. The best specifications per country are indicated with darker colors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

a relatively small country. In this case, a single factor is sufficient to summarize the cross-sectional variation in GVA.

As a general pattern, we find that the relationship between the number of factors and the nowcast performance is approximately U-shaped; the predictive performance first increases, and, if Q is set too large, declines again. This pattern is particularly evident for large- and medium-sized countries, such as Germany, Greece, Spain, France, Italy, and the Netherlands. This suggests that overfitting may occur if the number of factors is set too large, providing some evidence that our factor model (if well specified) is capable of efficiently summarizing the bulk of cross-regional variation. It is worth noting that the penalty in terms of predictive losses is usually slightly larger for underfitting specifications when compared with overfitting ones.

All results discussed so far have been based on country-specific loss measures. In the next step, we delve deeper into how nowcast performance varies with Q by considering cross-regional variation within a given country. Fig. 1 includes boxplots for each country. The lower and upper quantiles in each boxplot represent the variation in CRPS across NUTS2 regions, and the median is the unweighted median performance per country. The upper panel reports the CRPS relative to the historical mean benchmark, while the bottom panel plots the raw CRPS.

The figure yields several relevant insights. First, the U-shaped relationship between Q and nowcast performance is observed once again for large- and medium-sized countries. This is evidenced by the decrease in median CRPSs with increasing values of Q (before a slight deterioration), as well as the downward shift in the full cross-sectional distribution of CRPSs.

Second, the cross-sectional dispersion in forecast performance declines with increasing values of Q. This implies that more factors better accommodate local dynamics, resulting in more homogeneous forecast performance

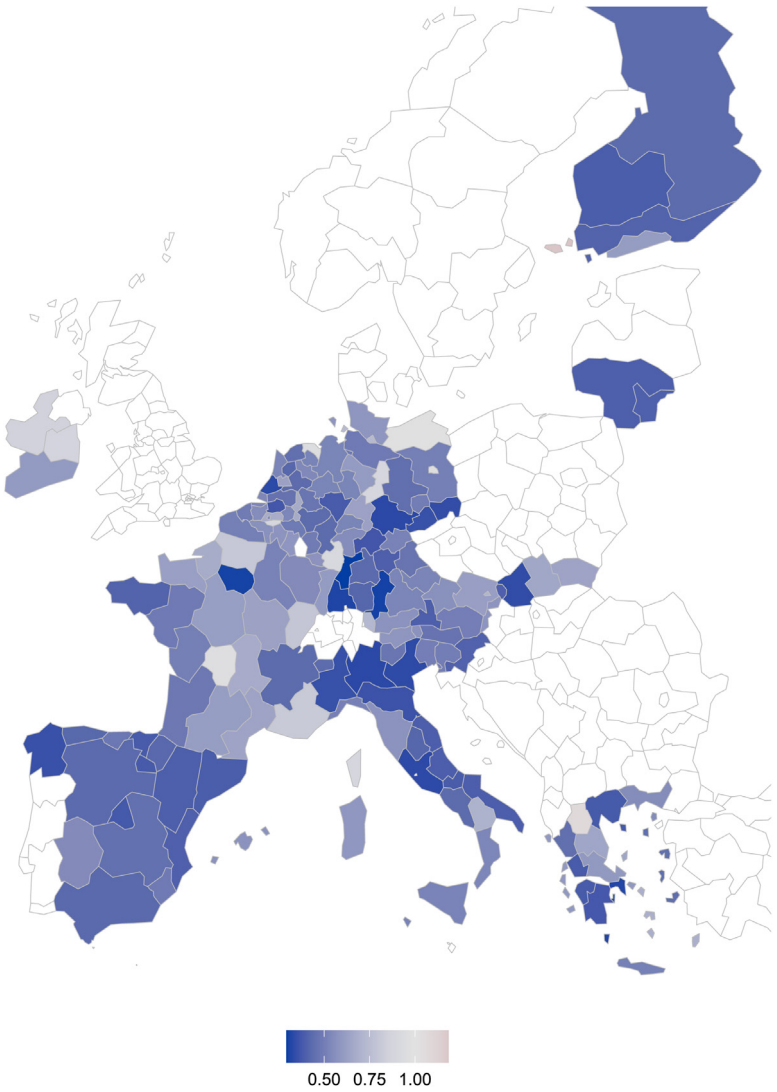


Fig. 2. Average CRPS across hold-out periods (2016–2023) by NUTS2 for MF-DFM models. The CRPS is reported relative to the one of the HM: values below 1 (in blue) indicate a better performance than the benchmark. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for larger values of Q. However, it is worth stressing that this reduction in cross-sectional variance is not linear; we find a significant reduction of variance when the number of factors is low, but the effect of increasing the number of factors, if Q is large, is often negligible.

Third, adding too many factors hurts nowcast performance not only on aggregate but also for most regions under consideration. Recall that the fully-specified model (which sets $Q = N$ or $Q = 12$) approximates the MF-VAR of Koop et al. (2020), in particular if the number of NUTS2 regions is below 12. We find that more parsimonious models with fewer factors often achieve gains, with Austria, Belgium, Finland, and Ireland being notable exceptions.

The boxplots provide only a rough gauge of the overall cross-regional variation. To shed further light on how well our model (with an optimal number of factors) works in particular regions, we present a map of average nowcasting gains in Fig. 2. Regions shown in blue indicate a better performance of the proposed model compared to the historical mean benchmark. The proposed models outperform the benchmark in 98% of the 161 regions, with an average gain of around 46% across regions. Only in three out of 161 regions does the historical mean improve upon our proposed specification. These regions are located in Finland, Germany, and Greece, suggesting that this relatively weaker performance is not country-specific and that there is no apparent systematic pattern. There are also several regions where improvements vis-á-vis the historical mean are particularly pronounced. These are mainly located in Italy, Lithuania, and Spain, with gains exceeding 50%.

Considering average predictive performance over the holdout period masks information about whether our model performs differently over distinct points in time.

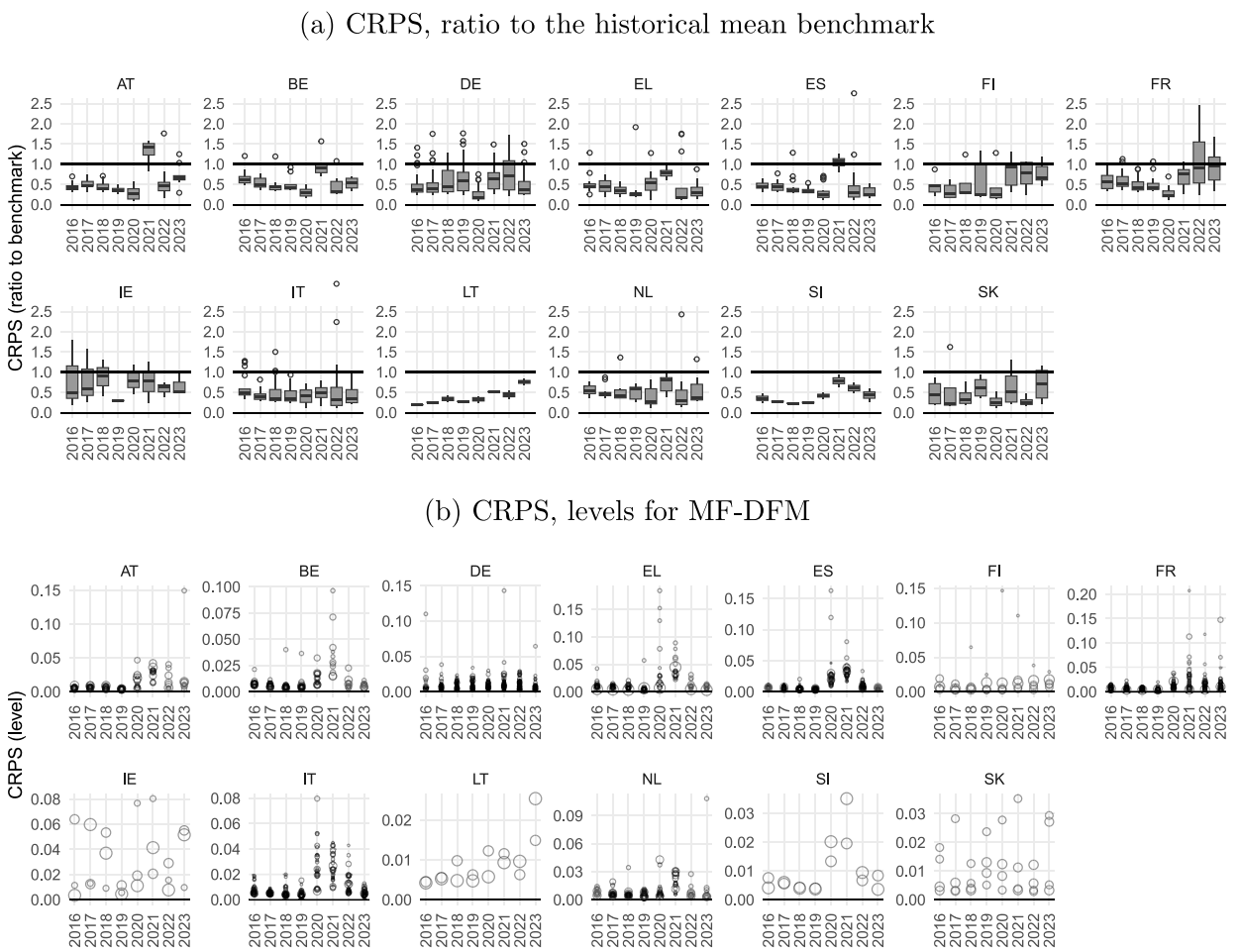


Fig. 3. Boxplots of relative regional CRPS across out-of-sample years for the best performing MF-DFM models by country in the upper panel (a) and raw CRPS losses per region in the lower panel (b). The CRPSs in panel (a) are shown as ratios to the historical mean nowcast (values below 1 signal improvement and vice versa), and circles are outliers. The size of the circles in panel (b) indicates the relative weight of the region in national GVA.

To gain a better understanding of whether the good performance of our model is driven by a few periods in which the historical mean benchmark performs poorly or whether the performance is stable and consistently better throughout the holdout, we consider the relative CRPSs over time. These are in the form of boxplots, which measure cross-sectional variation, as shown in Fig. 3. Again, we select the respective model specification based on the average CRPS performance over time.

The figure reveals that the forecast performance relative to the benchmark is generally stable over time, with some distinct patterns during the COVID years. In most countries, the onset of the pandemic and the initial downturn in 2020 are captured much more accurately compared to the historical mean benchmark. Interestingly, we observe a weaker performance in absolute terms in the rebound of 2021. However, looking at the relative CRPS, we observe that only in Austria and Spain the proposed model is outperformed by the historical mean. In all other country-year combinations, the MF-DFM attains a significantly higher accuracy.

Considering the raw CRPS (which can be interpreted in the same scale as the data) reported in the bottom panel of Fig. 3, at least two aspects are worth noting. First, some regions are more challenging to predict than others, and average losses in certain countries exhibit heterogeneity. For instance, over the full holdout period, losses in Irish regions tend to be larger than those in all other countries. By contrast, raw CRPSs in Lithuania, Slovenia, and Slovakia are smaller in size. Second, the deterioration of nowcast accuracy during the height of the COVID-19 pandemic is clearly visible. Other than in Spain and Austria, this is particularly evident in Belgium, Greece, and Italy. It is noteworthy that the predictive losses stabilize to pre-COVID levels by 2022 in most countries, suggesting that our modeling framework is capable of adequately recovering from non-standard periods contained in the sample.

3.3. The pandemic through the rear-view mirror

High-frequency estimates of regional GVA can be an essential tool for policy formulation and ex post evaluation. This section serves two main purposes. First, we discuss full-sample estimates of regional GVA and illustrate some of the implications of our modeling approach

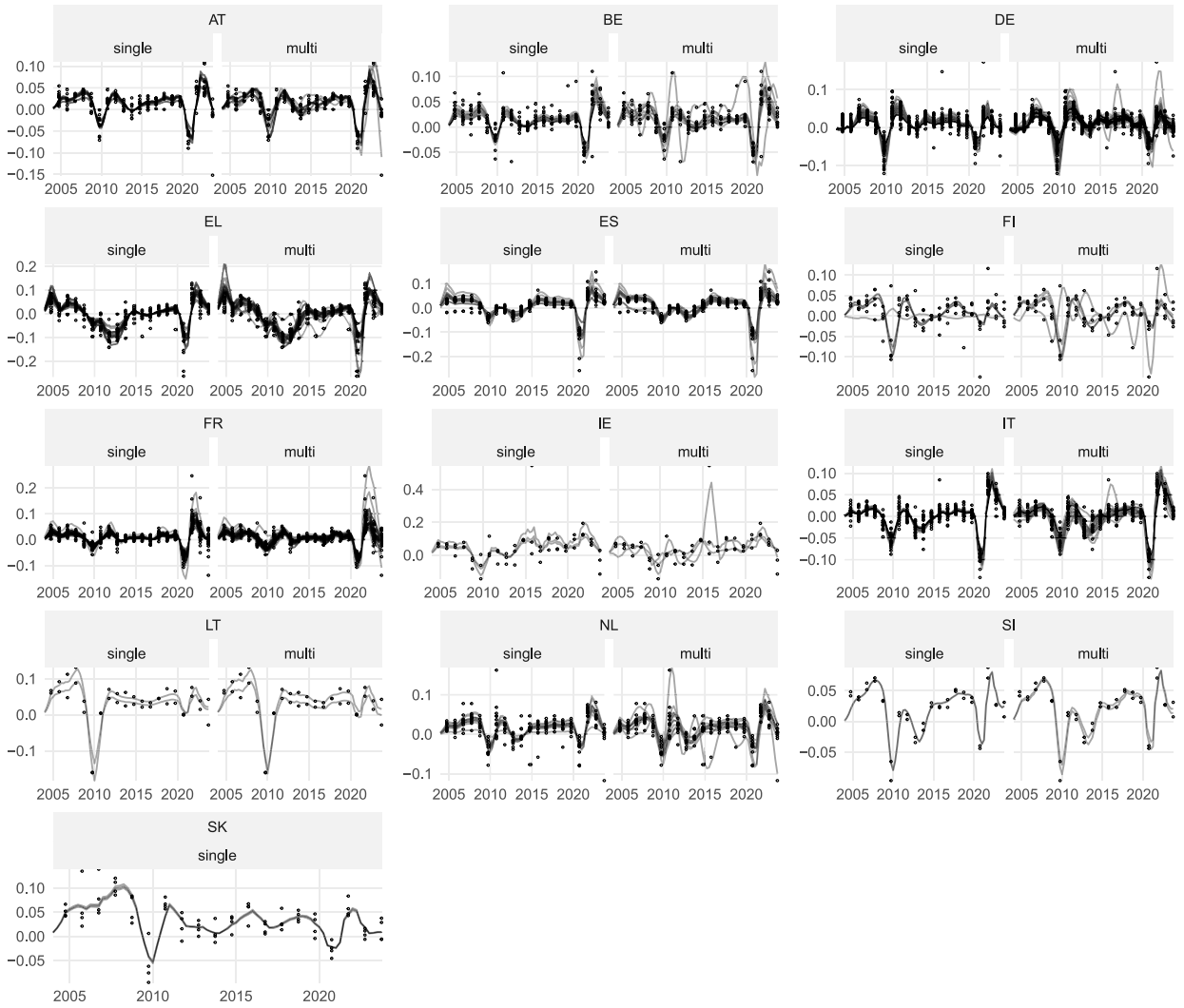


Fig. 4. Full-sample posterior median estimates of regional GVA growth (annualized quarterly measures), best-performing specification by country, multi- and single-factor variants. Circles are annual observations of the cross-section of regional GVA growth. Note that in SK, the best-performing model is a single-factor specification.

while also presenting other potentially useful outputs produced by our model. We achieve this by comparing single- and multi-factor versions of the best-performing model specification by country, as selected based on our results in Section 3.2, across time. Second, these results are complemented by a review of potential regional heterogeneities in the effects of the COVID-19 pandemic, which we assess using a small case study of the first few quarters of the 2020s, based on estimates from our full sample.

Full sample estimates. We present the posterior median estimate of our annualized quarterly regional GVA measures, $\boldsymbol{y}_t^{(a)}$, for the best-performing model by country in Fig. 4 in the form of a black line for each region. The circles are annual observations of the cross-section of regional GVA growth, i.e., $\boldsymbol{y}_t^*$. To illustrate the implications of underestimating the number of factors, we complement each of the country-specific panels with the corresponding single-factor model (keeping all other specification choices apart from the number of factors, such as priors or whether we use stochastic volatilities fixed). As noted in Table 1, in all countries, but Slovakia, a multi-factor model performs best. Additional results for the latent quarterly GVA growth measures, $\boldsymbol{y}_t$, can be found in Appendix A.

Overall, we find that business cycle dynamics within countries are often very similar, although the peaks and troughs may differ in terms of both timing and magnitude. Examples of such deviations are evident in Belgium, Finland, and the Netherlands. This provides evidence of the merits of our approach, as the framework is flexible enough to allow for region-specific idiosyncrasies when using multiple factors.

Another important finding is that the quarterly regional series tend to be correlated across countries. Given that we consider independent models for the regions

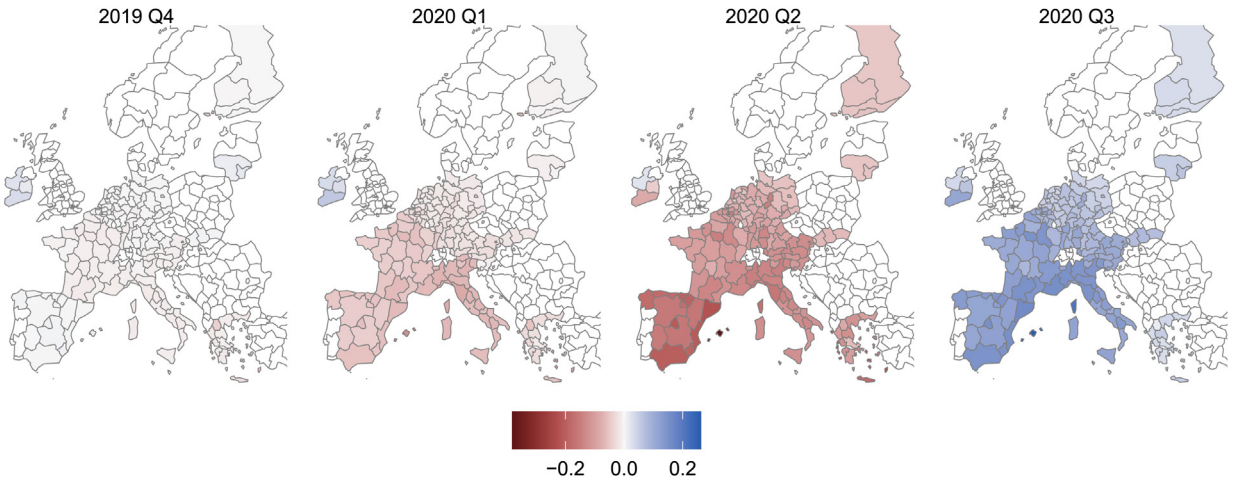


Fig. 5. Latent quarterly estimates of regional GVA growth in 2019 Q4 and 2020 Q1-3, full sample estimation. Negative values are represented in red, while positive values are represented in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of each country, we do not exploit this apparent empirical regularity. We again note that extensions of our framework, e.g., featuring all European regions at once subject to country-specific and area-wide factors, may be an interesting aspect for future research, but comes with several complications that we discussed earlier. A noteworthy exception to the co-movement across countries is Ireland. Irish regional output is affected by similar factors as national GDP measurement (e.g., the presence of a large number of US-multinational companies, see FitzGerald, Duffy, McQuinn, Byrne, & Morley, 2015).

Varying the number of factors has several implications for the resulting annualized quarterly series. By design, the single-factor case results in the regional GVA series being proportional to one another, since they are scaled versions of the same underlying latent quarterly state. The model thus has a harder time balancing the fit across all cross-sectional units, and the actual observations in many cases are comparatively far from the estimated regional series. Compared to the multi-factor model, the measurement errors (and their variances) are significantly larger in this specification, as the error term compensates for the model's underfitting.

We now focus on regional heterogeneities in GVA dynamics during the first few quarters of 2020, that is, during the period that was perhaps the most turbulent of the COVID-19 pandemic. Fig. 5 shows the posterior median of the quarterly regional growth rates, $\boldsymbol{y}_t$, in the form of a map. As an example, we investigate the behavior of the estimated high-frequency GVA in correspondence with the outbreak of the COVID-19 pandemic in Europe (see Carvalho, Krammer, & Iwasaki, 2021 for a timeline of the COVID-19 evolution). Fig. 5 plots the maps of quarterly GVA growth in the last quarter of 2019 and the first three quarters of 2020 estimated using our full sample.

The plot for 2019Q4 shows the pre-pandemic situation. While COVID-19 was already spreading at the end of 2019, its impact on economic activities became apparent only from the first quarter of the following year. The majority of European regions experienced mildly positive output growth. The first major governmental restrictions to limit the spread of the virus in Europe were implemented at the end of February 2020 in Northern Italy. Similar measures across the continent soon followed. The second panel of Fig. 5 shows the estimated GVA growth in 2020Q1. Compared to the previous quarter, we observe a clear downturn in economic conditions across Europe, except for Ireland.

Moving to the second quarter of 2020, the situation deteriorated in all regions in our sample. Europe is now the epicenter of the COVID-19 pandemic, with the highest number of reported cases and deaths worldwide, accompanied by mobility restrictions and lockdowns.[14] The largest GVA contractions are observed in some coastal and insular regions in Spain, Greece, and Italy. These territories are characterized by a strong dependence on the tourism and hospitality industries, which were among the most impacted by the pandemic (Plzáková & Smeral, 2022). By 2020Q3, the rebound of output is clearly visible and reflects the patterns observed in most national output time series.[15]

Pseudo real-time estimates. The previous discussions relate to ex-post considerations of regional output dynamics with full-sample information. For policymakers, it is not only evaluations after the fact, but also real-time monitoring of regional heterogeneities, that is important. For example, in cases of budgetary constraints, these estimates can help allocate resources efficiently and promote cohesive policies across European territories in a more timely manner than with annual snapshots of regional performance, which are released with a two-year delay.

For the following discussions, we compare the predictions from our pseudo real-time out-of-sample scheme (which respects the release calendar) with the full sample

14 See this World Health Organization statement.

15 The data set of latent and annualized quarterly GVA estimates is available alongside the `R` code to estimate the proposed model and replicate the paper's results.

estimates we mentioned before. The latter serves as our current best estimate of the state of annualized quarterly regional GVA. In contrast, the former can be used to assess how new information impacts backcasts and nowcasts at different points in time. As is common in the related literature, we refer to these distinct information sets across time when predictions are made as "vintages". However, we note again that these indeed reflect missing data, but not data revisions.

Fig. 6 depicts selected predictive distributions for nowcasts and backcasts made at different points in time. The black crosses represent realized values, while the black dot-dashed lines indicate the 68 percent posterior of the estimated regional GVA using full sample information (from 2001 to 2023, encompassing all annual regional observations). The colored shaded bands and lines refer to the 68 percent credible set and median of annualized quarterly GVA estimates. These are obtained in pseudo-real time as discussed in Section 3.2 (e.g., the "2020" vintages use annual regional information until 2018, but include national quarterly data until 2020Q4, etc.). Because the final two regional observations are missing for each vintage, we indicate both the backcast ($T-4$, empty circle) and nowcast (T, filled circle). As a full year passes, we obtain a new annual regional observation alongside four quarters of national information. That is, the previous year's backcast becomes a nowcast, and so on. Crucially, this new information may also affect estimates of earlier quarters for which annual regional information is already available; therefore, we always show the complete sequence of predictions until the respective nowcasted quarter.

Our estimations produce outputs for each region in our sample. These results are rather high-dimensional, and it is infeasible to present them all. This is why we pick a set of illustrative examples that feature typical characteristics found across most regions. Specifically, we focus on the "Big Four" European economies, Germany (DE), Italy (IT), France (FR), and Spain (ES); within these countries, we select a "Largest" (region with maximum share) and a "Median" (region with median share) region based on their share in national output. These selected NUTS regions are shown in Fig. 6.

These charts reveal several important findings. Overall, in line with our discussion of the predictive losses in Section 3.2, our model does a reasonable job of tracking movements in regional output growth. While in some regions (e.g., DE21, DE27, and ITC4), both the nowcast and backcast are close to the realized value, in the other regions, the magnitude of the downturn in 2020 was underestimated. Similarly, the subsequent rebound in 2021 was underestimated for most regions, apart from FR10, with varying margins relative to the actual realizations. We note again that this is just a small subset of regions. As suggested by our full predictive evaluations above, these patterns still reflect substantial improvements relative to simpler benchmarks on average. Our model also appears to be relatively robust to the inclusion of pandemic observations, as evidenced by the accurate nowcasts for 2022 and 2023. As new information enters the model, we observe a decline in predictive uncertainty and in several cases, revisions to earlier estimates. These occur mainly when new annual regional information is observed. Interestingly, as the nowcast turns into a backcast (i.e., when comparing the backcast and nowcast for the same reference quarter for successive vintages), adjustments in predictions are typically modest. This finding is also suggestive of the robustness of our proposed approach to potentially unusual dynamics in national quarterly information.

3.4. State equation diagnostics and parameter estimates

Macroeconomic data from the COVID-19 pandemic is unusual for several reasons. Among other things, the measurements of many variables are far from their usual means, which may affect model estimation. In this context, it is crucial to evaluate whether the model's parameters are significantly impacted when the respective information is present in the sample. As discussed above, the performance of our model in terms of predictive losses remains comparatively stable throughout the full hold-out period, and this stability is also observed in most countries, as well as in the nowcasts and backcasts. We now analyze some of the parameters that govern the state equation of our MF-DFM. We use the eigenvalues of the companion-form matrix of the VAR in (3) as a summary of how much the VAR coefficients are affected over all vintages. Specifically, we compute the posterior median of the real and imaginary parts of the respective eigenvalue with the maximum modulus. These estimates are shown on the unit circle in panel (a) of Fig. 7 for the best-performing model specification across countries. We complement these results, which relate to the dynamics of the latent and observed states, with a summary statistic of the variance of the state equation. Here, we compute the log-determinant of the state innovation matrix, and show the 68 percent posterior credible set alongside the median estimate, in panel (b) of Fig. 7.

We start our discussion with the dynamic properties of the respective state equations. Considering panel (a) of Fig. 7, some heterogeneity across countries and vintages is clearly visible. Specifically, while there is virtually no variation across time for Belgium, Germany, Finland, or the Netherlands, including the national COVID-19 Pandemic output observations, it affects the stationarity properties of the state equation in other countries. For instance, in Austria and Spain, there is a shift of the maximum modulus eigenvalues towards the stationarity boundary. In Slovenia and Slovakia, the real parts are affected, likely due to a different eigenvalue having the maximum modulus. And sometimes, such as in Spain or France, the imaginary part appears to increase slightly in absolute value, which may be explained by the oscillating behavior of national output in 2020Q2 (downturn) and Q3 (rebound).

In panel (b), the figure shows $\log(\det(\boldsymbol{H}))$, a rough gauge of the overall dispersion or uncertainty across the dimensions of the underlying multivariate distribution. If the best-performing model features stochastic volatilities, we use the unconditional mean over time to calculate this metric. Overall, we find that uncertainty tends to increase in most countries by 2020, driven by national quarterly

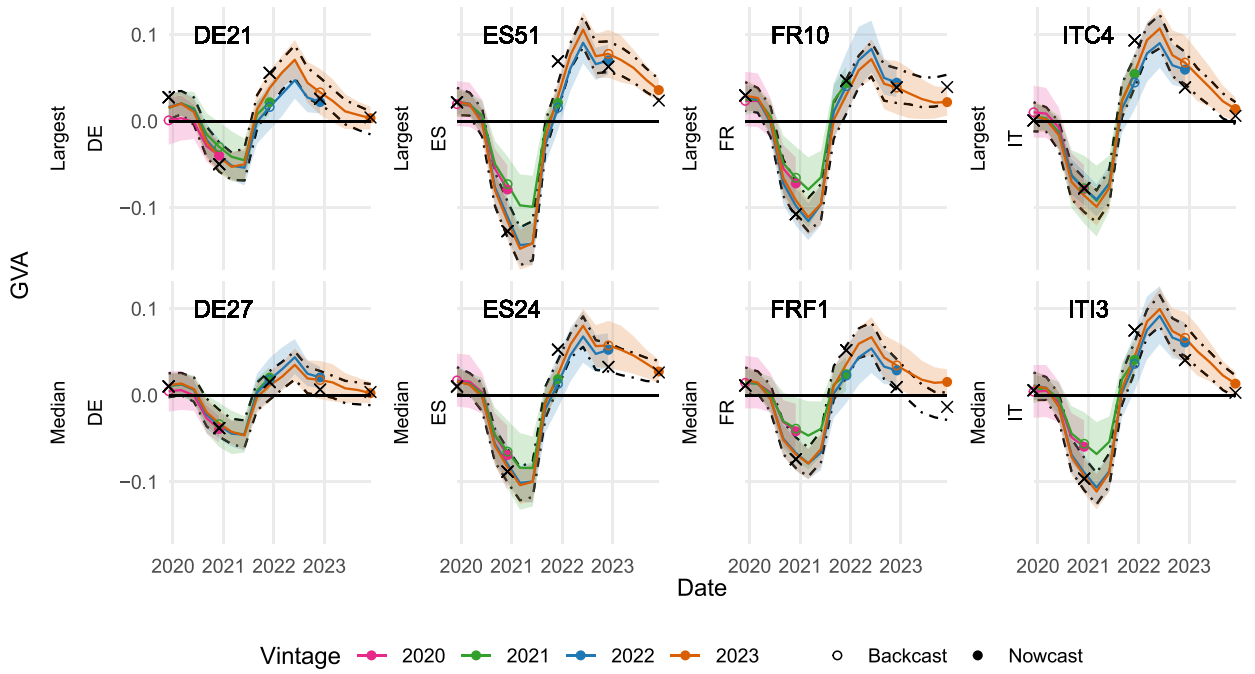


Fig. 6. Selected predictive distributions for nowcasts and backcasts made at different points in time. Black crosses are realized values, the dot-dashed band marks the 68 percent posterior of the estimated annualized regional GVA when using full sample information (2001–2023); x-axis labels refer to the beginning of each indicated year. Colored shaded bands and lines refer to the 68 percent credible set of estimates in pseudo-real time (e.g., the "2020" vintages use annual regional information up to 2018, etc.). The "Largest" and "Median" regions per country (the four biggest economies in Europe) are selected based on their share in national output. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

observations (at this point, regional annual information is not yet included in the information set). Appendix A contains several additional results for unconditional variances at the regional level, which are largely consistent with these findings.

Summarizing, we find that our framework is responsive to new information, both in terms of the parameters' estimates and the resulting predictions. At the same time, our approach appears to be robust to the inclusion of non-standard economic episodes, thereby delivering a relatively consistent out-of-sample performance.

4. Conclusions and future research

Using regional data typically gives rise to (at least) two challenges. First, key series, such as GVA (as a measure of output), are often only available on a yearly frequency and are released with significant lags. For example, in our European dataset, there are instances of delays exceeding two years. Second, estimating joint regional models implies that the number of time series to process becomes large. We address both issues by proposing an MF-DFM model.

Our model flexibly combines national GVA data, available at the quarterly frequency, with regional GVA series that are only available at an annual frequency. We introduce temporal and cross-sectional restrictions to interpolate the yearly series and thus obtain quarterly measures of regional GVA. We thoroughly investigate the choice of the number of endogenous factors, the implications of choices about priors, and leverage independent estimates for the different European countries, which are characterized by varying numbers of regions, as well as other sources of heterogeneity. In a nowcasting exercise, we show that our approach yields precise density forecasts.

The basic model developed in this paper can be extended along further directions. For example, one shortcoming is that we model the countries separately. In principle, pooling all series together and thus estimating a huge dimensional factor model would be feasible. Another possible extension of our model is to allow for time-varying parameters in the state and/or the observation equation. Finally, from a practitioner's perspective, it is worth emphasizing that our model can also serve as a tool for conducting structural inference at the regional level.

CRediT authorship contribution statement

Luca Barbaglia: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lorenzo Frattarolo:** Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Niko Hauzenberger:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Dominik Hirschbühl:** Writing – review & editing, Writing

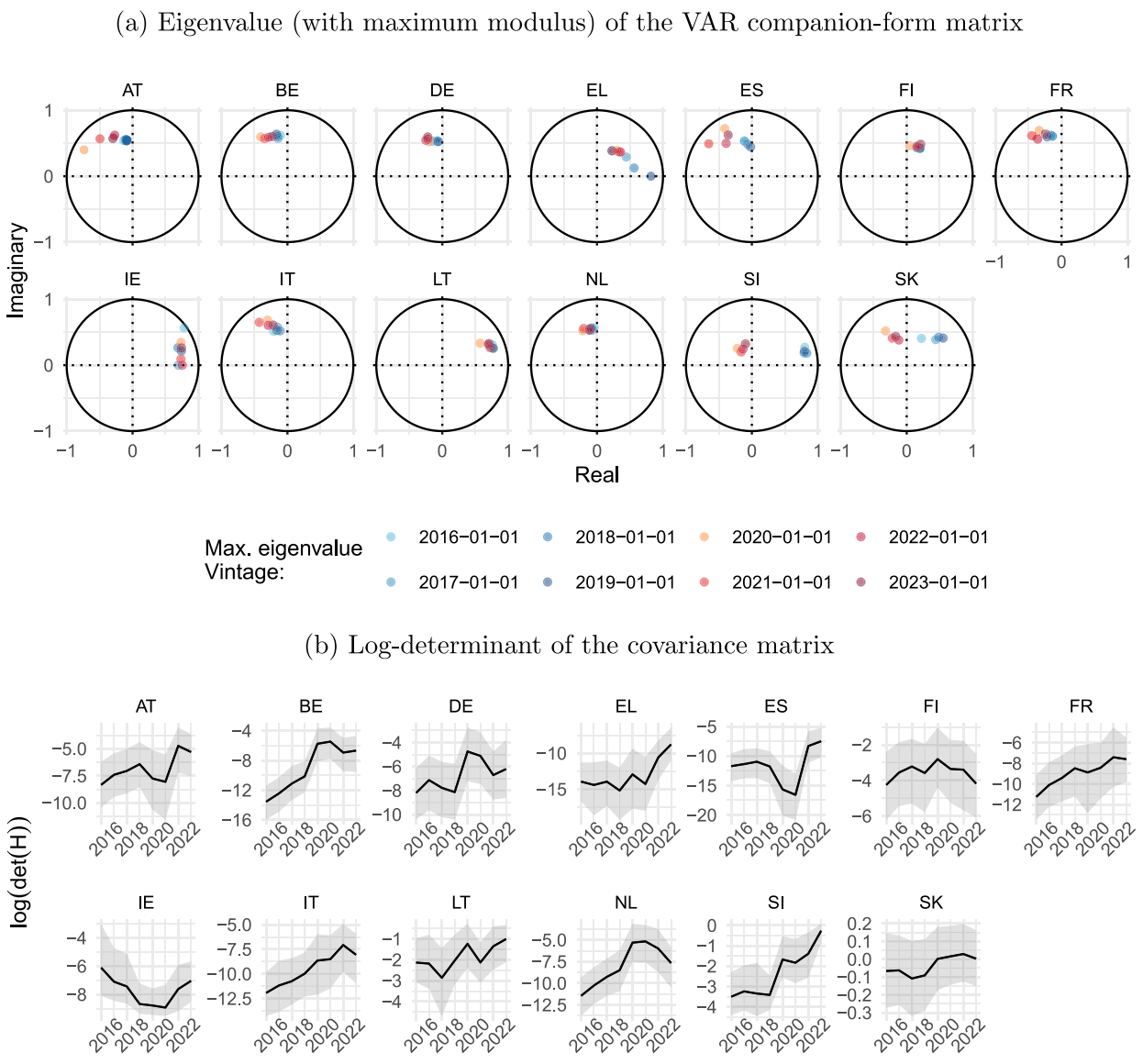


Fig. 7. Summary statistics for state equation parameters, best specification by country. Posterior median (upper panel) and 68 percent credible set alongside median (lower panel).

– original draft, Validation, Supervision, Formal analysis, Conceptualization. **Florian Huber:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Luca Onorante:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Michael Pfarrhofer:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Luca Tiozzo Pezzoli:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.09.006.

Data and code availability

The codes and data to reproduce the results in the paper "Nowcasting economic activity in European regions

using a mixed-frequency dynamic factor model" are available at https://github.com/lucabarbaglia/RegionalNowcasting_replication_package_IJoF.

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